EPFL
École Polytechnique Fédérale de Lausanne
School of Computer and Communication Sciences

COM-480 — Data Visualization · 2025

Is Lausanne's housing stock evolving to match its *changing social fabric?*

Milestone 2 — Project Report

Lens 1 · Who lives here | **Lens 2 · How people live** | **Lens 3 · What housing offers**

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https://lausanne-evolution-team.github.io/lausanne-evolution/website/

Is Lausanne's housing stock evolving to match its changing social fabric?

1. Project Goal

Lausanne is a city in transformation. Over four decades it has grown significantly, absorbed large waves of international residents — today nearly half the population is foreign-born — and seen its household composition shift markedly toward smaller, single-person living arrangements.

Central question: *Is Lausanne's housing supply adapting to the way its population is changing?*

The visualization explores this through **three interlocking lenses**. This structure allows us to progressively build the argument: from demographic change, to its impact on living patterns, and finally to its implications for housing supply. **Lens 1** — who lives in Lausanne (demographic change since 1979); **Lens 2** — how people live and work (household composition and employment); **Lens 3** — what housing offers (room and surface distribution). The narrative culminates in a **core gap chart** that quantifies the mismatch between demand for small dwellings and their actual supply. Target audience: the general public, as well as urban planners and policymakers interested in housing and demographic change.

2. Sketches

LENS 1 — WHO LIVES IN LAUSANNE Population by sex, origin & age · 1979–2025

Stacked area chart

Swiss vs foreign share · 1979–2025

Tool: D3.js
Lecture: D3.js (9 Mar) · Interactions (16 Mar)

MVP

Choropleth map

Foreign % by district · year slider

Tool: D3.js + GeoJSON
Lecture: Practical maps (20 Apr) [future]

MVP

Population pyramid

Age groups by sex · animated slider

Tool: D3.js
Lecture: D3.js (9 Mar) · Interactions (16 Mar)

EXTRA

LENS 2 — HOW PEOPLE LIVE Households by size · Employment by sex · 2012–2025

Dual line chart

1-person HH% vs mean HH size · 2012–2025

Tool: D3.js
Lecture: D3.js (9 Mar) · Interactions (16 Mar)

MVP

Connected dot chart

Female headcount % vs FTE% · part-time gap

Tool: D3.js
Lecture: Marks & channels (23 Mar) · D3.js (9 Mar)

MVP

Scatter plot

Foreign% vs single-HH% per district · 2024

Tool: D3.js
Lecture: Marks & channels (23 Mar)

EXTRA

LENS 3 — WHAT HOUSING OFFERS Dwellings by rooms & surface · 2011–2025

Gap area — CORE

1-person HH% vs 1–2 room% · gap = shortage

Tool: D3.js
Lecture: D3.js (9 Mar) · Interactions (16 Mar)

CORE

Grouped bar chart

Room dist. vs HH size dist. · supply vs demand

Tool: D3.js
Lecture: Marks & channels (23 Mar) · D3.js (9 Mar)

MVP

Surface stacked area

Dwelling size distribution · 2010–2024

Tool: D3.js
Lecture: D3.js (9 Mar) · Interactions (16 Mar)

EXTRA

CORE Must work — directly answers the central research question MVP Required for a complete project EXTRA Enhances depth, safely droppable

3. Tools & Lectures

Tool	Used for	COM-480 Lecture
D3.js	All charts — scales, axes, areas, stacks, lines, bars, dots	**D3.js** (9 Mar) · **More interactive D3** (16 Mar)
D3.js + GeoJSON	Choropleth map · district colour encoding	**Practical maps** (20 Apr) *[future]*
D3.js transitions	Line draw, bar grow, area fade-in animations	**Interactions** (16 Mar) · **More interactive D3** (16 Mar)
Intersection Observer	Scroll-triggered chart reveals · storytelling	**Storytelling** (18 May) *[future]*
Python · pandas	Data preprocessing — wide→long, derived ratios	**Data** (9 Mar)
HTML / CSS	Page layout, navigation, grid, tooltips	**Basic web development** (16 Feb)
JavaScript	Data loading, DOM manipulation, interaction logic	**Javascript Part 1 & 2** (23 Feb, 2 Mar)

4. Goal Breakdown — Core Visualization & Independent Pieces

Core visualization (MVP)

The **gap area chart** is the minimum viable product. It plots the share of 1-person households (demand) against the share of 1–2 room dwellings (supply) from 2012 to 2024, with the area between the two lines filled in red. This single chart directly answers the central question. All other pieces add narrative context but cannot break the project if absent.

Piece	What it delivers	Charts
Piece 1 · Lens 1	Demographic context — establishes who lives in Lausanne and how that has changed	Stacked area · Choropleth · Pyramid
Piece 2 · Lens 2	Demand characterisation — shows the rise of small households and employment patterns	Dual line · Connected dot
Piece 3 · Lens 3 CORE	Supply vs demand — directly answers whether housing matches social change	**Gap area chart** · Grouped bar
Piece 4 · Narrative	Makes the conclusion explicit and memorable	Annotations · Insight text · Scroll transitions

This gap represents a structural mismatch: demand for small dwellings is growing faster than supply, indicating a persistent shortage in Lausanne's housing stock.

5. Extra Ideas (can be dropped without endangering the project)

- Real GeoJSON district map (replace colour grid)
- Scrollytelling strip with narrative callouts
- District click-through filter across all charts
- Radar chart — district profile vs city average
- Waffle/unit chart — 37k HH vs 35k small apts
- Bump chart — district rankings over time
- Population pyramid extended to district level
- Mobile-responsive layout

6. Functional Prototype

An initial website is running with the basic skeleton of the visualization. The site is structured as a single scrollable page with three lens sections and a sticky navigation bar. Each section contains D3-powered chart containers sized to their target dimensions. The stacked area chart (Lens 1), dual line chart (Lens 2), and gap area chart (Lens 3) are already rendered with real data from Statistique Lausanne. All charts include hover tooltips, and the population pyramid includes an animated year slider (1979–2024).

Prototype: https://lausanne-evolution-team.github.io/lausanne-evolution/website/